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Arduino UNO R4 Minima ADC Resolution

Learn how to change the ADC resolution on the UNO R4 Minima.

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In this tutorial you will learn how to change the analog-to-digital converter (ADC) on an **Arduino UNO R4 Minima** board. By default, the resolution is set to 10-bit, which can be updated to both 12-bit (0-4096) and 14-bit (0-16383) resolutions for improved accuracy on analog readings.

Goals

The goals of this tutorials are:

- Update the ADC resolution to 12/14-bit.

Hardware & Software Needed

- Arduino IDE (online or offline)
- Arduino UNO R4 Minima
- UNO R4 Board Package

Help

Analog-to-Digital Converter (ADC)

An analog-to-digital converter (ADC) transforms an analog signal to a digital one. The standard resolution on Arduino boards is set to 10-bit (0-1023). The UNO R4 Minima supports up to 14-bit resolutions, which can provide a more precise value from analog signals.

To update the resolution, you will only need to use the analogReadResolution() command.

To use it, simply include it in your `setup()`, and use `analogRead()` to retrieve a value from an analog pin.

```
void setup(){
  analogReadResolution(
}

void loop(){
  int reading = analogR
}
```

Summary

This short tutorial shows how to update the resolution for your ADC, a new feature available on the UNO R4 Minima board.

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